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Highly Tough, Notch-Insensitive, and Fast Self-Healing Thermoplastic Polyurethane Elastomers by Tailored Soft Segment Design

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ABSTRACT

Achieving toughness, notch insensitivity, and self-healing within a single thermoplastic polyurethane (TPU) system is challenging. Hard domains formed by hydrogen-bonded urethane linkages provide mechanical strength but limit the chain mobility needed for healing, while soft domains enable elasticity but are more vulnerable to damage. We propose a soft-segment modification strategy that incorporates α,ω-hydroxyl end-functionalized polyacrylates into conventional TPU structures. Unlike our previous systems based on linear-alkyl acrylates (i.e., *n*-butyl acrylate), introducing branched-alkyl acrylic soft segments (i.e., 2-ethylhexyl acrylate) generate side-chain architectures whose segmental dynamics respond more sensitively to temperature. This enhanced thermal responsiveness promotes thermally activated chain mobility and more uniform stress distribution, all without compromising the mechanical contribution of the hard domains. As a result, the acrylic TPUs display high toughness (186 $MJ \cdot m^{-3}$), stable notch performance (fracture energy >197 $kJ \cdot m^{-2}$ at elongations above 900%), and practical self-healing (~90% recovery at 45°C within 3 h). These findings highlight that branched polyacrylate–based soft-segment engineering offers a balanced pathway to mechanically robust and adaptable TPU elastomers.

1 | Introduction

Thermoplastic polyurethane (TPU) elastomers are block copolymers characterized by a distinctive microphase-separated structure composed of hydrogen-bonded hard domains dispersed within flexible soft segments [1]. The hard segments, derived from short-chain diols and diisocyanates, function as physical crosslinks that impart mechanical strength and structural integrity, while the soft segments, formed from long-chain polyols such as polyethers or polyesters, provide flexibility and elasticity (Figure 1a) [2–4]. This dual-phase architecture grants TPU elastomers an exceptional balance of elasticity, toughness, and processability [5, 6], enabling their widespread use in flexible components such as seals [7], gaskets [8], and footwear [9].

Despite their excellent mechanical properties, TPU elastomers often face structural limitations under severe or repetitive loading. Their segmented morphology, while responsible for excellent mechanical performance, is susceptible to damage accumulation

Changhoon Yu and Hongdeok Kim contributed equally to this work.

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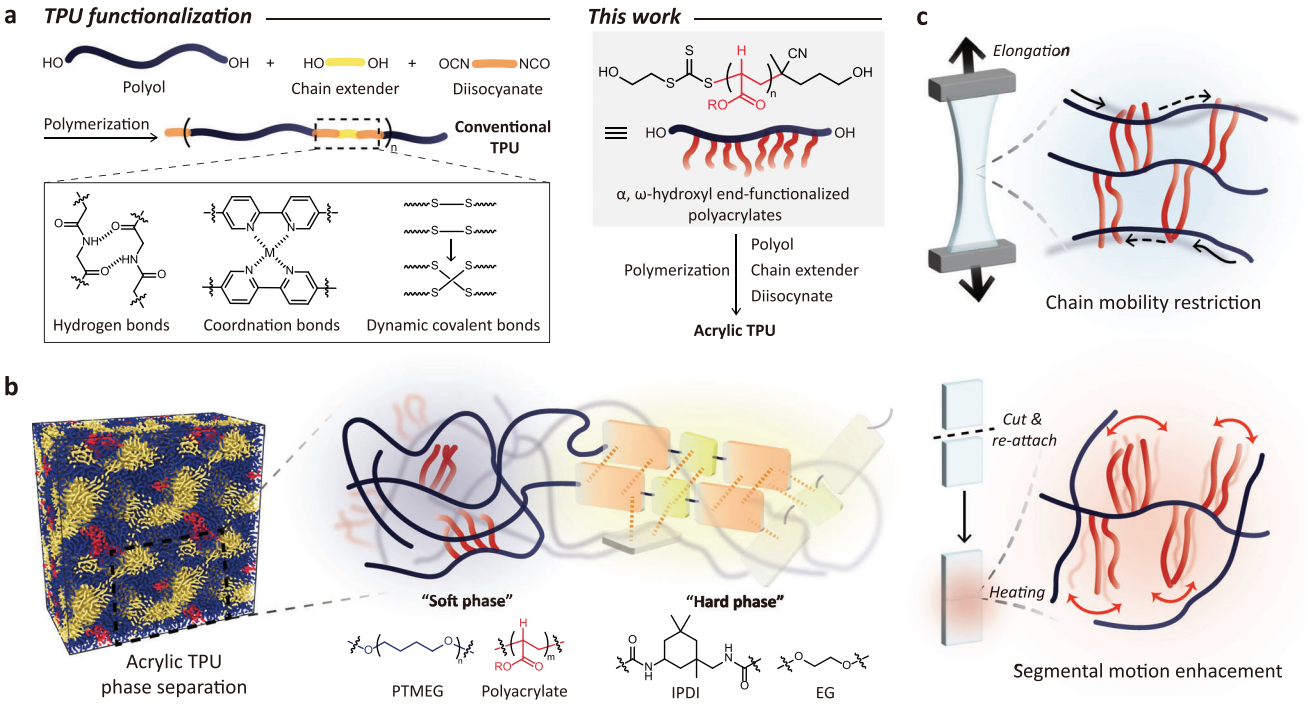


FIGURE 1 | (a) Preparation method of conventional TPU elastomer using polyol, chain extender, and diisocyanate giving functionalities into hard segment (left). Acrylic TPU elastomer composition by incorporating α,ω-hydroxyl end-functionalized polyacrylates as soft segment (right). (b) Molecular dynamic simulation results in phase separation of acrylic TPU, exhibiting polyacrylate embedded into soft segment. (c) Role of polyacrylates with TPU elastomers showing restricted cavitation formation during deformation as well as enhanced segmental motion under elevated temperatures.

through crack initiation and propagation under cyclic or severe mechanical loads [10]. Such deterioration leads to premature failure and shortened service lifetimes, posing significant concerns for applications requiring continuous operation and durability [11–13]. In particular, wearable and biomedical applications subject TPU elastomers to harsh and repetitive mechanical environments—characterized by bending, stretching, and abrasion—necessitating materials with high notch resistance and rapid self-healing capabilities [14–19]. Consequently, integrating multifunctionality into TPU elastomers, combining high toughness, notch-insensitivity, and rapid self-healing under mild conditions, has emerged as a challenge [20].

The challenges in TPU elastomers stem from the interplay between their microphase-separated morphology and failure mechanisms. Cracks initiate at defects and propagate through soft and hard domains—soft segments cavitate while hard domains concentrate stress, driving crack growth [21–23]. Self-healing relies on thermally activated chain mobility and reversible bonds, often requiring elevated temperature [24–26]. Structural approaches embed stiff sacrificial skeletons [27, 28] or form interlaced supramolecular networks to stabilize strain fields and dissipate energy, enhancing toughness and notch resistance [29–31]. Dynamic chemistries such as metal–ligand coordination [32, 33] and disulfide exchange [16, 34–37] enable rapid repair near room temperature with minimal thermal input. However, networks with high dynamic mobility heal well but lack strength, while reinforced hard domains increase toughness but restrict molecular rearrangements needed for healing [38, 39]. Consequently, achieving TPU elastomers that unite mechanical resilience and rapid self-healing under mild conditions remains an important, unresolved goal [40].

Addressing the long-standing trade-off between mechanical robustness and dynamic recovery requires a fresh molecular design perspective as most conventional strategies have focused primarily on hard-segment modification. Soft-segment engineering provides an alternative route by enabling control over viscoelastic relaxation, cavitation resistance, and strain distribution while preserving the structural role of hard domains [41, 42]. Introducing flexible, energy-dissipative motifs into the soft phase allows TPU elastomers to maintain more uniform strain, delay hard-domain dissociation, and facilitate chain rearrangement under mild thermal conditions—factors essential for both durability and healing [43]. Guided by these considerations and prior progress, we incorporated α,ω-hydroxyl end-functionalized polyacrylates as soft segments (Figure 1a) [44].

Our earlier work established a synthetic platform for embedding defect-free α,ω-hydroxyl end-functionalized polyacrylates into TPU matrices, but the resulting materials showed only modest overall improvements, and the mechanisms governing deformation and healing were not fully understood. Notably, despite achieving reasonably high toughness, the materials displayed little to no notch resistance, indicating that the soft-segment design did not yet provide the strain redistribution or cavitation suppression needed to manage stress concentrations. More broadly, the previous study did not clarify how polyacrylates influence cavitation behavior, load transfer, or hard-domain dynamics, nor did it demonstrate the concurrent realization of high toughness, notch resistance, and rapid self-healing within a single TPU system.

In this study, we advance our earlier work by developing acrylic TPU elastomers that use precisely engineered polyacrylate

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soft segments to balance mechanical robustness with dynamic recovery. Molecular dynamics (MD) simulations and experiments show that these polyacrylates preferentially reside within the soft domains (Figure 1b), where their brush-like side chains help dissipate stress, reduce cavitation, promote more uniform deformation, and increase the temperature sensitivity of segmental motion (Figure 1c). Guided by these insights, we refined the soft-segment design from poly(*n*-butyl acrylate) to poly(2-ethylhexyl acrylate), which leads to higher toughness (186 $MJ \cdot m^{-3}$) and improved self-healing (~90% efficiency at 45°C within 3 h) compared to our previous system. Notably, the new elastomers also exhibit strong notch insensitivity, achieving a fracture energy of 197.5 $kJ \cdot m^{-2}$ at elongations above 900% in notched samples—performance that was not attainable with the earlier poly(*n*-butyl acrylate)-based TPUs. Overall, these findings show that polyacrylate-based soft-segment engineering offers an effective approach to combine mechanical resilience with dynamic recovery, providing a practical design direction for future functional TPU elastomers.

2 | Results and Discussion

2.1 | Acrylic TPU Elastomers Preparation

To develop acrylic TPU elastomers with enhanced performance, it is essential to clarify the role of polyacrylates within the TPU matrix. Our previous work demonstrated that α,ω-hydroxyl end-functionalized polyacrylates can be incorporated as part of the soft segment, enabling defect-free chain integration and yielding modest improvements in mechanical and healing properties [44]. However, that study did not resolve how acrylate-based soft segments influence microphase separation, cavitation behavior, or deformation mechanisms.

Building on this foundation, we systematically introduced α,ω-hydroxyl end-functionalized polyacrylates into the TPU backbone at controlled loadings and evaluated their effects on mechanical performance, phase separation, and viscoelastic response. This systematic approach establishes direct structure–property relationships and provides deeper insight into how soft-segment engineering contributes to the simultaneous enhancement of toughness, notch resistance, and self-healing in TPU elastomers.

To achieve this objective, α,ω-hydroxyl end-functionalized polyacrylates were synthesized using visible-light photoiniferter reversible addition–fragmentation chain-transfer (RAFT) polymerization. As shown previously, this method yields α,ω-hydroxyl end-functionalized poly(*n*-butyl acrylate) (ph-PBA) with near-Poisson dispersity (Mn ≈ 3,500 g/mol, $Đ = 1.03$) and high end-group fidelity (Figures S1 and S2), enabling seamless incorporation into TPU networks. A series of acrylic TPUs were then fabricated by substituting 0.5–10 mol% of polytetramethylene ether glycol (PTMEG, Mn ~1,000 g/mol) with ph-PBA to obtain $PTMEG_{1-x}$–ph-PBA_x–EG_2–IPU_3 (Figure 2a), where x denotes the molar fraction of ph-PBA relative to PTMEG. TPU polymerization was performed in toluene at 70°C overnight using dibutyltin dilaurate (DBTDL) as the catalyst, with isophorone diisocyanate (IPDI) as the diisocyanate and ethylene glycol (EG) as the chain extender. This procedure yielded high-molecular-weight elastomers (Mn = 50,000–90,000 g/mol), with complete urethane formation confirmed by FT-IR and ^{1}H NMR (Figures S3 and S4 and Table S1).

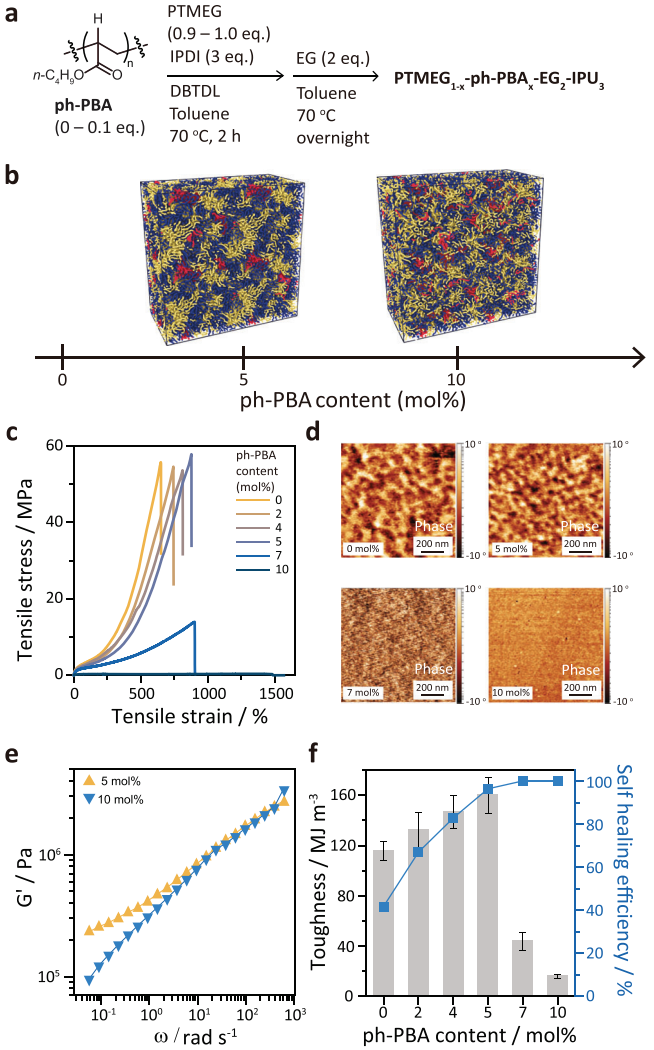


FIGURE 2 | (a) Schematic of the synthetic route and formulation protocol for preparing $PTMEG_{1-x}$–ph-PBA_x–EG_2–IPU_3, showing systematic substitution of conventional PTMEG with ph-PBA at 0–10 mol%. (b) Visualizations of atomistic and AFM phase images illustrating the morphological transition from clear phase-separated domains (≤5 mol% ph-PBA) to a nearly homogeneous morphology (10 mol% ph-PBA). (c) Representative tensile stress–strain curves for TPUs with varied ph-PBA content, showing mechanical enhancement at low substitution and reduced toughness when phase separation is lost. (d) AFM phase images for TPUs at different ph-PBA contents, highlighting the shift from separated microphase domains to homogeneous morphology as ph-PBA increases. (e) Frequency sweep results of storage modulus (G') for TPUs with 5 and 10 mol% ph-PBA. (f) Corresponding plot of toughness and self-healing efficiency as a function of ph-PBA content. Error bars represent standard deviations from five samples.

In contrast to our earlier work, the present study systematically varies polyacrylate content and directly probes their structural and mechanical roles, thereby establishing previously unresolved connections between soft-segment composition and macroscopic properties. This acrylic TPU series serves as the platform for the mechanistic investigations that follow, including strain-resolved

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atomic force microscope (AFM), rheology, and molecular dynamics simulations.

2.2 | Characteristics of Acrylic TPU Elastomers by Varying Amounts of Polyacrylates

To systematically investigate the properties of the synthesized acrylic TPU elastomers, free-standing films with a thickness of approximately 200 μm were prepared (see Supporting Information for detailed methods). Since the elastomeric performance of TPU elastomers is predominantly governed by their phase-separated morphology, we first characterized their microphase separation using AFM. In the neat TPU $PTMEG_1-EG_2-IPU_3$ formulation, AFM images revealed distinct microphase-separated domains, with bright and dark regions corresponding to the hard and soft segments, respectively [36]. When ph-PBA substitution was at or below 5 mol%, the phase-separated morphology was largely maintained, aligning with the observed enhancement in mechanical properties (Figure 2b,d). However, upon increasing the ph-PBA content beyond this threshold, domain boundaries became increasingly diffused. Specifically, at 10 mol% ph-PBA ($PTMEG_{0.9}-ph\text{-}PBA_{0.1}-EG_2-IPU_3$), AFM images indicated a nearly homogeneous morphology (Figure 2d). This transition is likely attributed to hydrogen bonding interactions between acrylate carbonyl groups and urethane segments, which enhance thermodynamic miscibility but diminish phase separation.

Mechanical properties were then assessed via uniaxial tensile testing (Figures 2c and S5 and Table S2). Incorporation of up to 5 mol% ph-PBA preserved the tensile strength while significantly increasing elongation at break, resulting in notable toughness improvement (~38%). This enhancement is presumably related to delayed microvoid formation under large strain and strain-hardening behavior [31, 32], corroborated by amplitude sweep rheological tests demonstrating an expanded linear viscoelastic region (LVER) and slower destruction of phase-separated domains (details provided in subsequent sections) (Figure S6). To further elucidate the effects of ph-PBA incorporation on the viscoelastic behavior, frequency sweep measurements were conducted at room temperature (Figures 2e and S7). TPU samples containing up to 5 mol% ph-PBA exhibited comparable storage (G′) and loss (G″) moduli, indicating balanced viscoelasticity. Beyond this concentration, the storage modulus decreased sharply while the loss modulus increased, resulting in tan δ values exceeding unity—signifying a shift toward more viscous behavior.

The self-healing performance was evaluated by cutting the films, re-attaching them at 60°C for 2 h, and subsequently performing uniaxial tensile testing (Figures 2f and S8). Healing efficiency was calculated as the ratio of the toughness after healing to that before damage (Table S2). Remarkably, ph-PBA contents up to 5 mol% yielded simultaneous improvements in toughness and healing efficiency, indicating that up to 5 mol% incorporation of ph-PBA effectively mitigates the conventional trade-off between mechanical robustness and healing. In contrast, at higher than 5 mol% ph-PBA levels, the loss in phase separation led to compromised mechanical properties; nonetheless, increased chain mobility significantly improved healing efficiency—approaching near-complete recovery, albeit in softer, less mechanically robust films. Additional insights were gained from frequency sweeps (1% strain) over varied temperatures, combined with time-temperature superposition (TTS) master curve analysis (Figure S9) and stress relaxation experiments at 60°C (Figure S10). These rheological evaluations confirmed that 5 mol% ph-PBA incorporation facilitates efficient chain mobility and network rearrangement while maintaining mechanical integrity. Conversely, higher ph-PBA loading resulted in rapid degradation of tensile strength and toughness, consistent with prior phase observations that TPU systems predominantly composed of ph-PBA lack sufficient elasticity.

2.3 | Deformation Behavior of Acrylic TPU Elastomers

Understanding the deformation mechanism of TPU elastomers is critical to rationalizing their enhanced mechanical behavior. The macroscopic response of TPU elastomers is governed by the interplay between flexible soft segments and hydrogen-bonded hard domains within their microphase-separated structure. At low strains, the soft segments elongate reversibly, accommodating deformation through entropy-driven chain extension, while the hard domains act as reinforcing physical crosslinks that restrict chain mobility and maintain elasticity [21]. As strain increases, hydrogen bonds within the hard domains begin to dissociate, enabling the hard segments to reorganize and gradually orient along the stretching direction [45]. This progressive dissociation and reordering underpins the characteristic strain-hardening response of TPUs: energy is first dissipated by bond rupture, followed by additional reinforcement as hard domains align and form ordered structures—manifesting as a nonlinear, hyperelastic behavior. However, if this ordering is excessively rapid or incomplete, local stress accumulates, which can trigger catastrophic failure by hard-domain collapse or fracture.

To directly probe these processes, phase-resolved AFM analyses were conducted under tensile deformation (Figure 3a). For the conventional $PTMEG_1-EG_2-IPU_3$ system, the sequence of hard-domain dissociation, reorganization, and strain-induced hardening was evident. Initial symmetrical phase distribution of soft and hard segments (corresponding to negative and positive phase angles, respectively) evolved under elongation, showing a peak shift toward the hard segment region as strain increased—indicating hard-segment dissociation, reordering, and formation of denser hydrogen bonds (Figure 3b, top) [45, 46]. While this rapid ordering initially provided strength, it also promoted premature hard-domain collapse, ultimately resulting in brittle fracture.

In contrast, the acrylic TPU ($PTMEG_{0.95}-ph\text{-}PBA_{0.05}-EG_2-IPU_3$) demonstrated a slower, more gradual domain reorganization (Figure 3b, bottom). This behavior is quantitatively supported by AFM phase-angle analysis (Table S3). The mean phase angle $\langle\phi\rangle$, which reflects the average phase shift, and the full width at half maximum (FWHM), which captures the heterogeneity of local deformation states, both increase with elongation for all TPU formulations. The acrylic TPU exhibits a smaller increase in $\langle\phi\rangle$ and a narrower phase distribution compared with $PTMEG_1-EG_2-IPU_3$ [47, 48]. Consistently, the lower phase reorganization slope ($\Delta\phi/\Delta\varepsilon_{fracture}$) demonstrates

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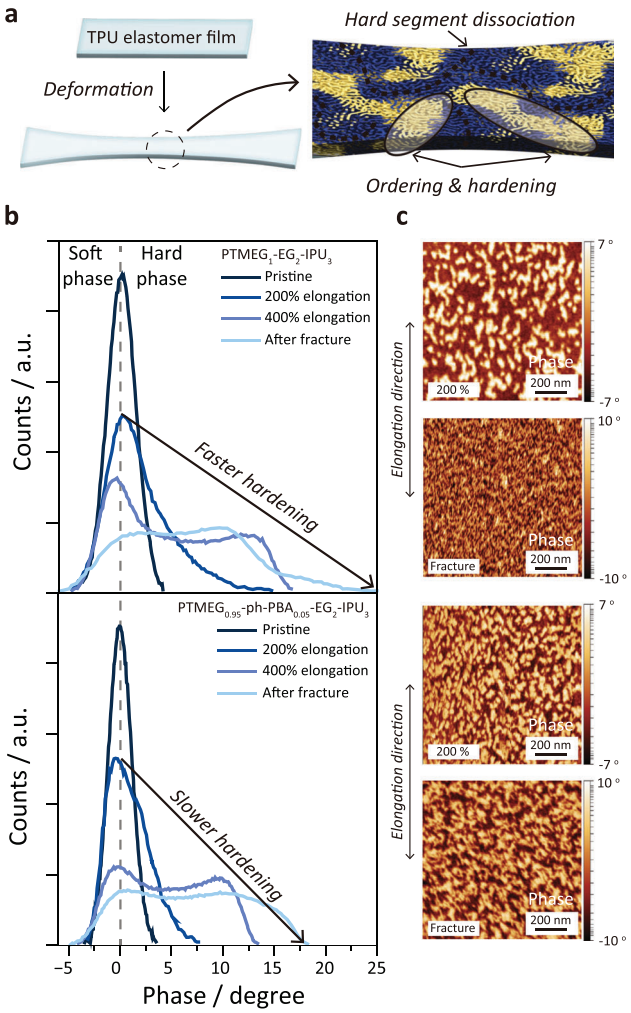


FIGURE 3 | (a) Schematic illustration of TPU deformation under tensile loading, highlighting hard-segment dissociation (red) and strain-induced ordering and hardening (blue) of microphase domains. (b) Phase distribution analyses during elongation for $PTMEG_1{-}EG_2{-}IPU_3$ (top) and $PTMEG_{0{\cdot}95}{-}ph\text{-}PBA_{0{\cdot}05}{-}EG_2{-}IPU_3$ (bottom). (c) AFM phase images of neat TPU (top) and acrylic TPU (bottom) at 200% elongation and fracture, confirming that acrylic TPU maintains larger, more interconnected hard domains under strain.

a reduced strain sensitivity of phase evolution, suggesting that incorporation of ph-PBA moderates strain-induced hard-domain reorganization and delays the onset of hardening, thereby extending the deformation window prior to fracture. AFM phase imaging corroborated these findings: while the neat TPU exhibited strongly oriented, fragmented domains after stretching and a reduction in hard domain size, the acrylic TPU maintained more interconnected and stable morphologies—even at large strains—with relatively larger hard domains (Figure 3c). Small-angle x-ray scattering (SAXS) measurements were also performed to examine strain-induced microstructural evolution (Figure S11). In the pristine state, both TPU elastomers exhibit characteristic domain spacings of approximately 5–6 nm. Upon tensile deformation, the scattering intensity gradually decreases and redistributes across the probed q-range, reflecting strain-induced distortion and partial disruption of hard-segment–rich domains, which reduce electron density contrast and broaden the distribution of characteristic domain spacing [10, 49]. Notably, compared with $PTMEG_1{-}EG_2{-}IPU_3$, $PTMEG_{0{\cdot}95}{-}ph\text{-}PBA_{0{\cdot}05}{-}EG_2{-}IPU_3$ exhibits a more gradual evolution of the SAXS profiles with increasing strain, indicating moderated disruption and reorganization of hard-segment–rich domains during deformation. Therefore, the ph-PBA group facilitates localized plastic deformation and stabilizes interfacial interactions, thereby suppressing catastrophic domain collapse and enhancing toughness.

These insights into deformation-induced microstructural evolution clarify how soft-segment engineering with ph-PBA delays hard-domain failure and enhances the resilience of the TPU network. As a result, these molecular-level modifications translate directly into improved extensibility, toughness, and self-healing capacity—critical features for advanced TPU materials intended for demanding mechanical applications.

2.4 | Molecular Mechanism Underlying Acrylic TPU Properties

To uncover the molecular-level origins of property enhancement in acrylic TPU elastomers, we employed MD simulations. Microstructural models of both neat TPU ($PTMEG_1{-}EG_2{-}IPU_3$) and acrylic TPU ($PTMEG_{0{\cdot}95}{-}ph\text{-}PBA_{0{\cdot}05}{-}EG_2{-}IPU_3$) were constructed (Figure 4a), with detailed modeling procedures provided in the Computational Details section of the Supporting Information. Briefly, TPU elastomers initially generated in an amorphous state underwent spontaneous phase separation due to thermodynamic incompatibility. At equilibrium, the ph-PBA segments preferentially localized within the soft domains (Figures 1b and S12), consistent with experimental observations that polyacrylate incorporation does not significantly disrupt the overall phase-separated morphology (Figure 2d).

Mechanical behavior was assessed through uniaxial tensile tests (Figure S13). Both $PTMEG_1{-}EG_2{-}IPU_3$ and $PTMEG_{0{\cdot}95}{-}ph\text{-}PBA_{0{\cdot}05}{-}EG_2{-}IPU_3$ exhibit elastic deformation at low strains, followed by significant hyperelasticity. Importantly, $PTMEG_1{-}EG_2{-}IPU_3$ exhibits abrupt failure at a true strain of approximately 1.3, while $PTMEG_{0{\cdot}95}{-}ph\text{-}PBA_{0{\cdot}05}{-}EG_2{-}IPU_3$ demonstrates no fracture within the examined strain range. Due to the difference in stiffness between each domain, fracture in $PTMEG_1{-}EG_2{-}IPU_3$ is predominantly initiated by localized cavitation and tearing within the soft domain (Figure 4a, left). In contrast, $PTMEG_{0{\cdot}95}{-}ph\text{-}PBA_{0{\cdot}05}{-}EG_2{-}IPU_3$ maintains a homogeneous strain distribution even under extreme mechanical deformation, thereby inhibiting localized cavitation (Figure 4a, right). The atomic strain distribution was analyzed using finite strain theory (Figure 4b). Prior to fracture, $PTMEG_1{-}EG_2{-}IPU_3$ exhibited amplified atomic strain within the soft domains. Because fracture is initiated when localized regions undergo severe deformation that impedes load transfer, this asymmetric strain distribution is detrimental to mechanical integrity. By contrast, the $PTMEG_{0{\cdot}95}{-}ph\text{-}PBA_{0{\cdot}05}{-}EG_2{-}IPU_3$ model showed a markedly reduced strain differential between domains, thereby facilitating more efficient load transfer under high strain conditions.

The origin of toughness enhancement was attributed to the brush-like chemical structure of the polyacrylates. Under large

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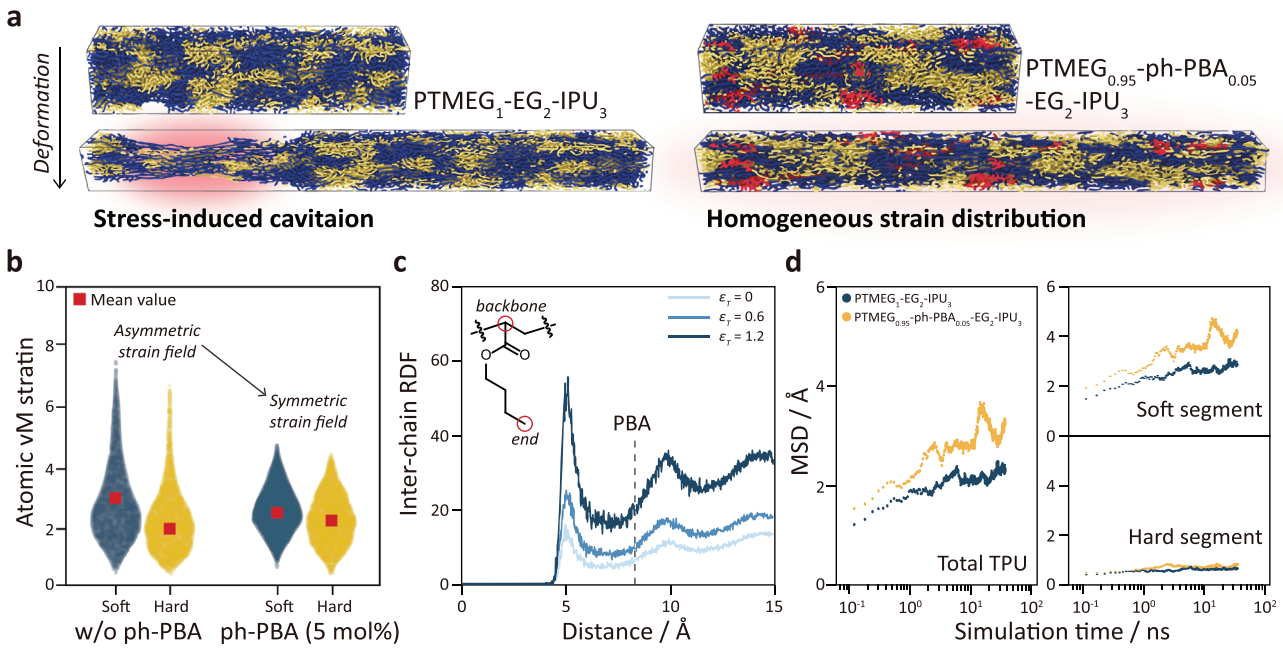


FIGURE 4 | (a) Molecular configuration of prepared TPU microstructures of TPUs during uniaxial deformations. The yellow and blue beads in the models represent the hard and soft segments, respectively. The polyacrylate soft segments in molecules (i.e., ph-PBA) are colored in red. (b) Atomic von Mises strain distribution of soft and hard domains immediately before fracture point ($\varepsilon_T = 1.3$). (c) Changes in intermolecular radial distribution function (RDF) during uniaxial tension. The vertical dotted lines represent the characteristic length (backbone-to-end) of ph-PBA. d) Molecular mobility evaluation of TPU components at 350 *K*.

deformation, the TPU backbone chains aligned along the loading direction, while the polyacrylate side chains aligned perpendicularly. This orthogonal arrangement introduced additional frictional resistance to intermolecular slippage of the soft segments, thereby preventing catastrophic tearing. Radial distribution function (RDF) analysis of the TPU backbone confirmed that intermolecular co-alignment intensified with applied strain (Figure 4c). Moreover, the characteristic length of the PBA segments enabled penetration into adjacent TPU backbones, further reinforcing structural integrity.

The thermal response of the ph-PBA segments was also investigated under conditions exceeding the glass transition temperature. Molecular mobility was quantified via mean square displacement (MSD) analysis (Figure 4d). The results demonstrated that incorporation of PBA significantly enhanced molecular mobility across the TPU matrix, with the most pronounced effect observed in the soft domains where PBA was localized. The brush-like architecture of PBA segments amplified thermal oscillations through their dangling ends, thereby promoting dynamic bond exchange and chain reorganization at elevated temperatures. This mechanism likely contributes to the improved self-healing efficiency of acrylic TPUs.

In summary, the incorporation of polyacrylate polyols into the soft segments of TPU provides two key functionalities (Figure 1c): (1) enhanced toughness, achieved by suppressing intermolecular slippage under extreme strain, and (2) improved molecular mobility, arising from amplified thermal oscillations of the dangling ends, which facilitates dynamic reorganization and self-healing.

2.5 | Enhanced Performance of Acrylic TPU via Acrylate Modification

Building on our mechanistic insights into the role of polyacrylates within TPU matrices, we further optimized the formulation to maximize both mechanical performance and self-healing efficiency. To balance toughness, notch insensitivity, and dynamic recovery, we introduced a bulkier polyacrylate, α,ω-hydroxyl-terminated poly(2-ethylhexyl acrylate) (ph-EHA), which possesses longer side chains and a lower glass transition temperature ($\sim$ -75°C) compared to ph-PBA (Figure 5a) [50]. This modification was anticipated to suppress cavitation, improve notch tolerance, and enhance healing efficiency by promoting segmental mobility and network adaptability.

The ph-EHA was synthesized using the photoiniferter RAFT polymerization strategy as ph-PBA. In the presence of the hydroxyl-functional iniferter CHHCA, 2-ethylhexyl acrylate underwent controlled polymerization at room temperature under 455 nm LED irradiation (200 mW cm^{-2}, 24 h, N_2 atmosphere) (Figure S14a), targeting a DP of 25. The resulting ph-EHA exhibited excellent structural control, with Mn = 4,500 g/mol, $Đ$ = 1.03 (Figure S14b), and high end-group fidelity confirmed by ^{1}H NMR (Figure S15), demonstrating suitability for TPU incorporation. Maintaining the previous TPU formulation (PTMEG, IPDI, and EG), we systematically substituted 3–5 mol% of PTMEG with ph-EHA, yielding $PTMEG_{1-x}$–ph-EHA_x–EG_2–IPU_3 (where x = molar fraction of ph-EHA) (Table S4). Among these, $PTMEG_{0.96}$–ph-$EHA_{0.04}$–EG_2–IPU_3 exhibited the most remarkable performance (Figure S16).

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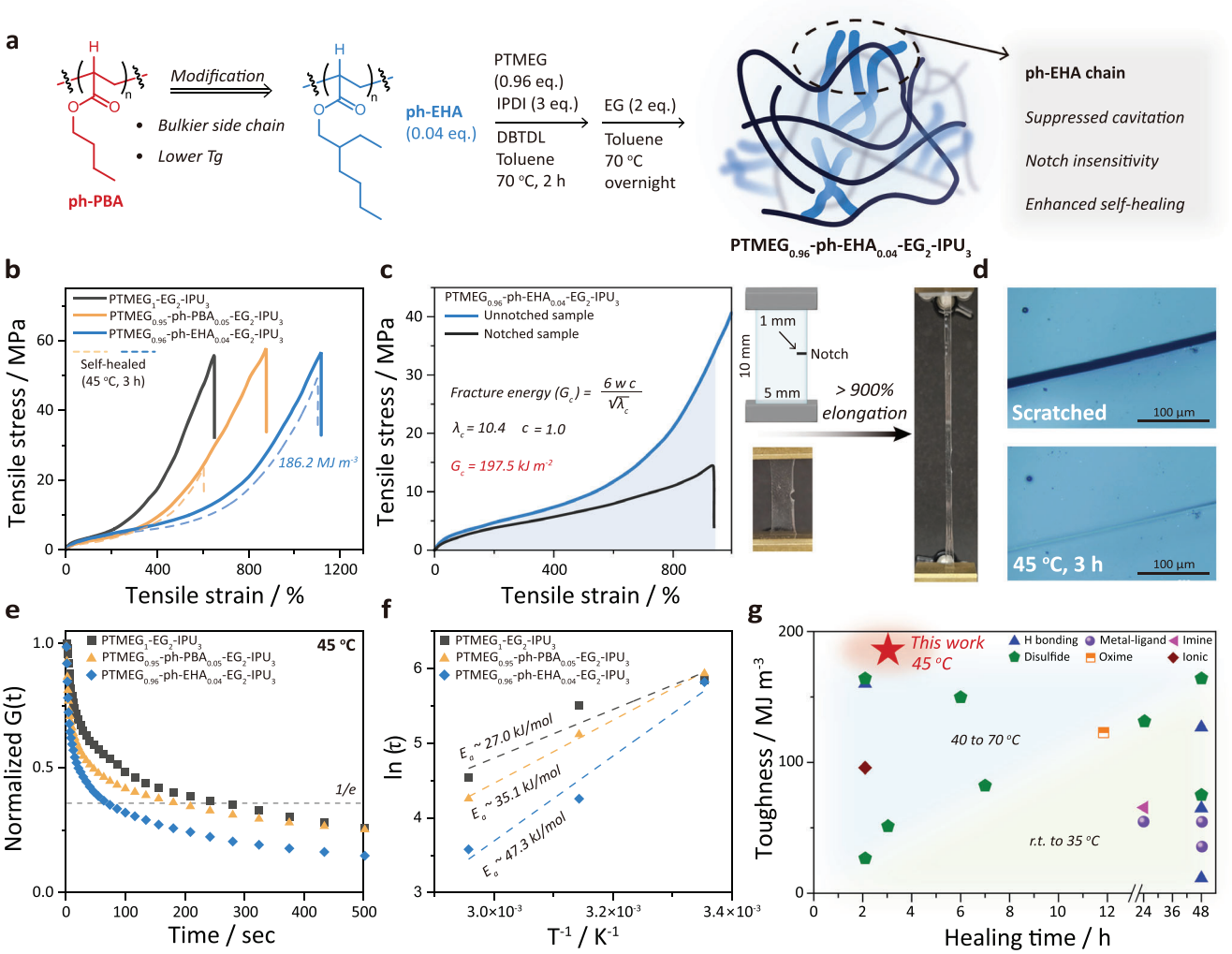


FIGURE 5 | (a) Molecular design concept: replacing ph-PBA with bulkier, lower-Tg α,ω-hydroxyl end-functionalized poly(2-ethylhexyl acrylate) (ph-EHA) soft segments to suppress cavitation, enhance notch insensitivity, and improve self-healing. (b) Stress–strain curves of prepared TPU elastomers, dashed lines indicate self-healed samples after 3 h at 45°C. (c) Tensile tests of notched specimens, and high fracture energy calculation for $PTMEG_{0\cdot96}$−ph-$EHA_{0\cdot04}$−EG_2−IPU_3. (d) Optical microscopy images of surface scratches (~20 μm width) before and after 3 h healing at 45°C. (e) Stress-relaxation tests at 45°C, revealing faster relaxation dynamics for ph-EHA–based TPU. (f) Arrhenius analysis of relaxation times based on stress-relaxation behavior under different temperatures. (g) Ashby plot of toughness versus healing time with reported self-healing elastomers, highlighting the unique combination of high toughness and rapid healing achieved by $PTMEG_{0\cdot96}$−ph-$EHA_{0\cdot04}$−EG_2−IPU_3 at moderate temperature (45°C, 3 h).

Uniaxial tensile tests yielded a toughness of 186.2 MJ·m^{-3} under pristine conditions (Figure 5b), while fracture tests confirmed exceptional notch insensitivity, with notched samples elongating beyond 900% and achieving a fracture energy (G_c) of 197.5 kJ·m^{-2} (Figure 5c). To further evaluate tear resistance properties, trouser tear tests were conducted and compared across the prepared TPU elastomers (Figure S17). The acrylic TPU elastomers display a more stable plateau force, indicating enhanced energy dissipation and tear resistance. Notably, $PTMEG_{0\cdot96}$−ph-$EHA_{0\cdot04}$−EG_2−IPU_3 maintains the most uniform and sustained tearing plateau over large displacements (~ 160 mm), accompanied by progressive deformation of the tear arms without catastrophic failure. Additionally, cyclic tensile tests were performed to assess the mechanical stability of the TPU elastomers under repeated deformation. As shown in Figure S18, all samples exhibit pronounced hysteresis during the first loading–unloading cycle, which is primarily attributed to the viscoelastic response of the soft segments and reversible rearrangement of hydrogen-bonded hard domains [21]. Although the acrylic-modified TPUs exhibit enhanced energy dissipation within the soft phase during deformation, they better maintain hard-segment domain integrity, resulting in reduced hysteresis upon repeated loading. Among them, $PTMEG_{0\cdot96}$−ph-$EHA_{0\cdot04}$−EG_2−IPU_3 exhibits the highest mechanical stability during repeated deformation. To correlate these macroscopic responses with microstructural evolution, phase-resolved AFM analyses were performed under the same deformation conditions (Figure S19). With increasing strain, $PTMEG_{0\cdot96}$−ph-$EHA_{0\cdot04}$−EG_2−IPU_3 exhibits a shift of the phase distribution toward higher phase angles, consistent with strain-induced domain reorganization. However, it shows the smallest increase in mean phase angle and the lowest phase reorganization slope ($\Delta\phi/\Delta\varepsilon_{fracture}$) (Table S3), indicating more uniform energy dissipation and better preservation of phase structure during deformation. MD simulations further support these observations. The von Mises strain analysis under tensile deformation reveals a more symmetric strain field, while RDF analysis

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indicates improved resistance to intermolecular slippage, due to the longer side chains of ph-EHA (Figure S20a,b). Collectively, these results highlight the superior stress-dissipation capability and deformation tolerance imparted by ph-EHA soft-segment modification.

Self-healing ability was assessed using cut-and-rejoin tensile tests. The $PTMEG_{0.96}$−ph-$EHA_{0.04}$−EG_2−IPU_3 formulation demonstrated an impressive ∼90% recovery in toughness after just 3 h of healing at 45°C, outperforming both neat and ph-PBA-based TPU samples (Figure 5b). Optical microscopy further corroborated these findings, showing nearly complete recovery of surface scratches (∼20 μm width) under the same healing conditions (Figure 5d), highlighting the efficient repair capability imparted by ph-EHA incorporation. The dynamic origin of this exceptional healing performance was investigated through stress-relaxation measurements. At 45°C, $PTMEG_{0.96}$−ph-$EHA_{0.04}$−EG_2−IPU_3 exhibited significantly faster relaxation compared to ph-PBA-containing counterparts (Figure 5e). Analysis of temperature-dependent relaxation times yielded activation energies (E_a) for dynamic bond exchange, with $PTMEG_{0.96}$−ph-$EHA_{0.04}$−EG_2−IPU_3 showing the highest E_a of 47.3 kJ mol^{-1} (Figures 5f and S21). This finding aligns with MD simulations, confirming that the bulky, low-glass-transition-temperature side chains of ph-EHA amplify thermal fluctuations, thereby facilitating dynamic rearrangements and accelerating healing kinetics (Figure S20c). Furthermore, thermal properties were investigated using temperature sweep tests. $PTMEG_{0.96}$−ph-$EHA_{0.04}$−EG_2−IPU_3 displayed a peak transition corresponding to the glass transition temperature (*T*g) near 20°C, characterized by a slightly reduced peak temperature and increased tan delta value during the transition compared to the $PTMEG_{0.95}$−ph-$PBA_{0.05}$−EG_2−IPU_3 formulation (Figure S22). This difference likely stems from the lower Tg of 2-ethylhexyl acrylate compared to *n*-butyl acrylate.

Finally, benchmarking against reported self-healing elastomers revealed that ph-EHA-based TPU elastomers offer an uncommon combination of high toughness (∼186 $MJ \cdot m^{-3}$) with rapid healing at moderate temperature (45°C, 3 h), comparing favorably with systems relying on hydrogen bonding, disulfide exchange, or other reversible chemistries (Figure 5g and Table S5). To the best of our knowledge, these materials represent a promising approach for achieving notch insensitivity, robust mechanical performance, and efficient self-healing within a single TPU platform. Collectively, these results suggest that polyacrylate-based soft-segment engineering may provide a versatile route for addressing the long-standing balance between mechanical robustness and dynamic recovery in TPU elastomers.

3 | Conclusions

This study demonstrates that soft-segment engineering through the incorporation of α,ω-hydroxyl end-functionalized polyacrylates can substantially enhance TPU elastomers by simultaneously improving toughness, notch insensitivity, and self-healing efficiency. In contrast to our earlier systems based on linear-alkyl acrylates, replacing them with branched-alkyl acrylate soft segments generates brush-like side-chain architectures that respond more sensitively to thermal stimuli and more effectively distribute stress. Molecular dynamics simulations reveal that these branched, brush-like motifs inhibit intermolecular slippage and promote thermally activated mobility, providing a molecular basis for the observed multifunctional enhancements. Experimental characterizations further confirm that controlled polyacrylate incorporation preserves microphase separation and delays hard-segment collapse, thereby stabilizing mechanical integrity under large deformation. As a result, the acrylic TPUs achieve record levels of toughness (∼186 $MJ \cdot m^{-3}$), fracture resistance (G_c = 197.5 $kJ \cdot m^{-2}$), and healing efficiency (∼90% at 45°C within 3 h). Overall, these findings highlight branched-acrylate soft-segment engineering as a versatile and scalable design strategy for high-performance TPU elastomers, while opening pathways toward adaptive, durable, and recyclable materials for advanced applications including flexible electronics, soft robotics, and biomedical devices.

Author Contributions

Changhoon Yu and Min Sang Kwon conceived the project. Changhoon Yu and Min Sang Kwon developed the system for mechanistic investigations of TPU elastomers and established the polymerization system design. Changhoon Yu performed most of the experiments, including mechanical testing, rheological analysis, and phase-separation characterization. Hongdeok Kim and Jinho Choi carried out the molecular dynamics simulations and contributed to the analysis of deformation mechanisms and the role of polyacrylate segments. Jinho Choi and Sunwu Song assisted in the development of the new soft-segment designs and in AFM measurements. All authors discussed the results and provided feedback on the manuscript. Changhoon Yu and Hongdeok Kim wrote the manuscript under supervision of Joonmyung Choi and Min Sang Kwon.

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Conflicts of Interest

Patent application (U.S. application number: 18/467,316) filed by M.S.K. and C.Y. from Seoul National University R&DB Foundation. Patent application covered preparation of α, ω-hydroxyl terminated polyacrylates and their application into TPU elastomers. The authors declare no other conflicts of interest.

Data Availability Statement

The authors declare that the data supporting the findings of this study are available within the paper and its Supporting Information.

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